

Digital Image Processing

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How many images do you generate every day?

Examples:

- Unlocking your phone using Face ID
- Taking selfies
- Instagram filters
- Google Maps
- CCTV cameras
- MRI scans
- Satellite weather images
- Drone inspection
- Autonomous vehicles

Almost every modern AI system **starts with an image**.



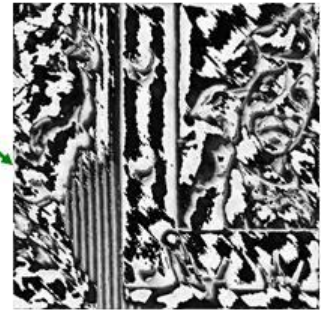
Original Image

Mesham
Example
with a low
pass filter

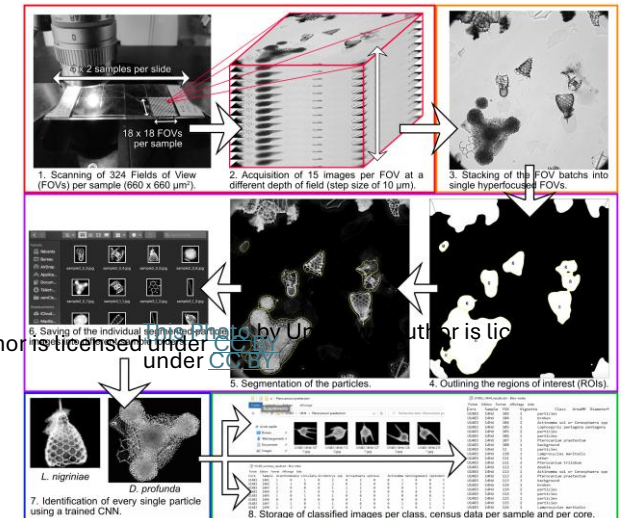


Result of low pass filter

Mesham
Example
with a high
pass filter



Result of high pass filter



What is an Image?

Imagine looking outside your classroom.

You see

- Trees
- Buildings
- Cars
- People
- Your eyes capture **continuous light**.
- A camera converts this into numbers.
- Those numbers form a **digital image**.



Real Example of Image Processing based Object Detection

Original selfie

↓

Increase brightness

↓

Remove noise

↓

Detect face

↓

Recognize person

↓

Unlock phone

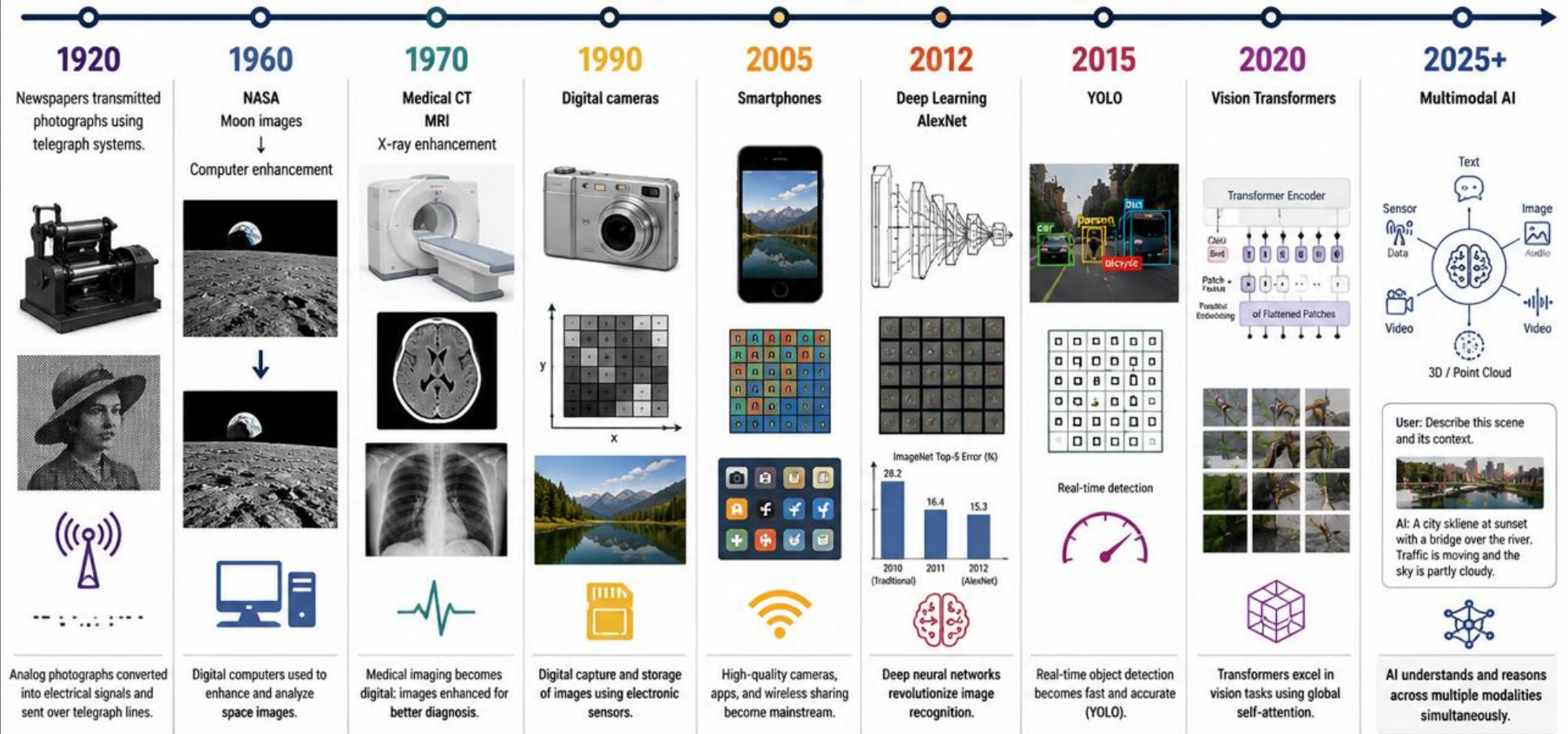
Notice

Only the first few steps are **Digital Image Processing**, The final recognition step belongs to **Computer Vision**.

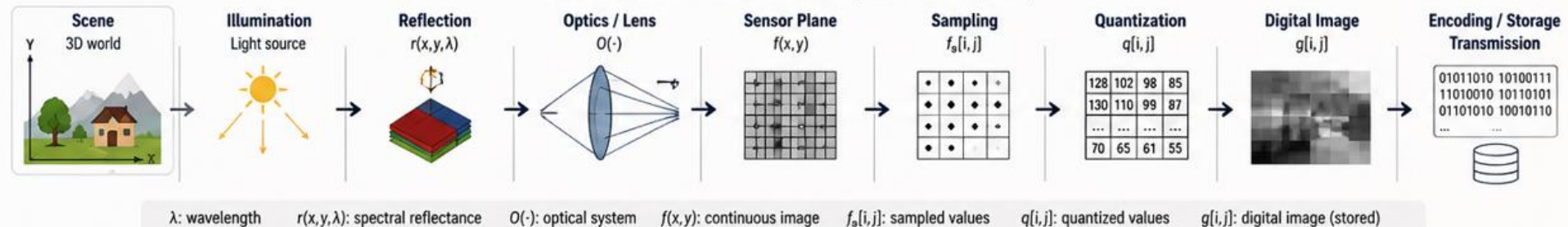
Digital Image Processing is the science of improving or transforming images using a computer.

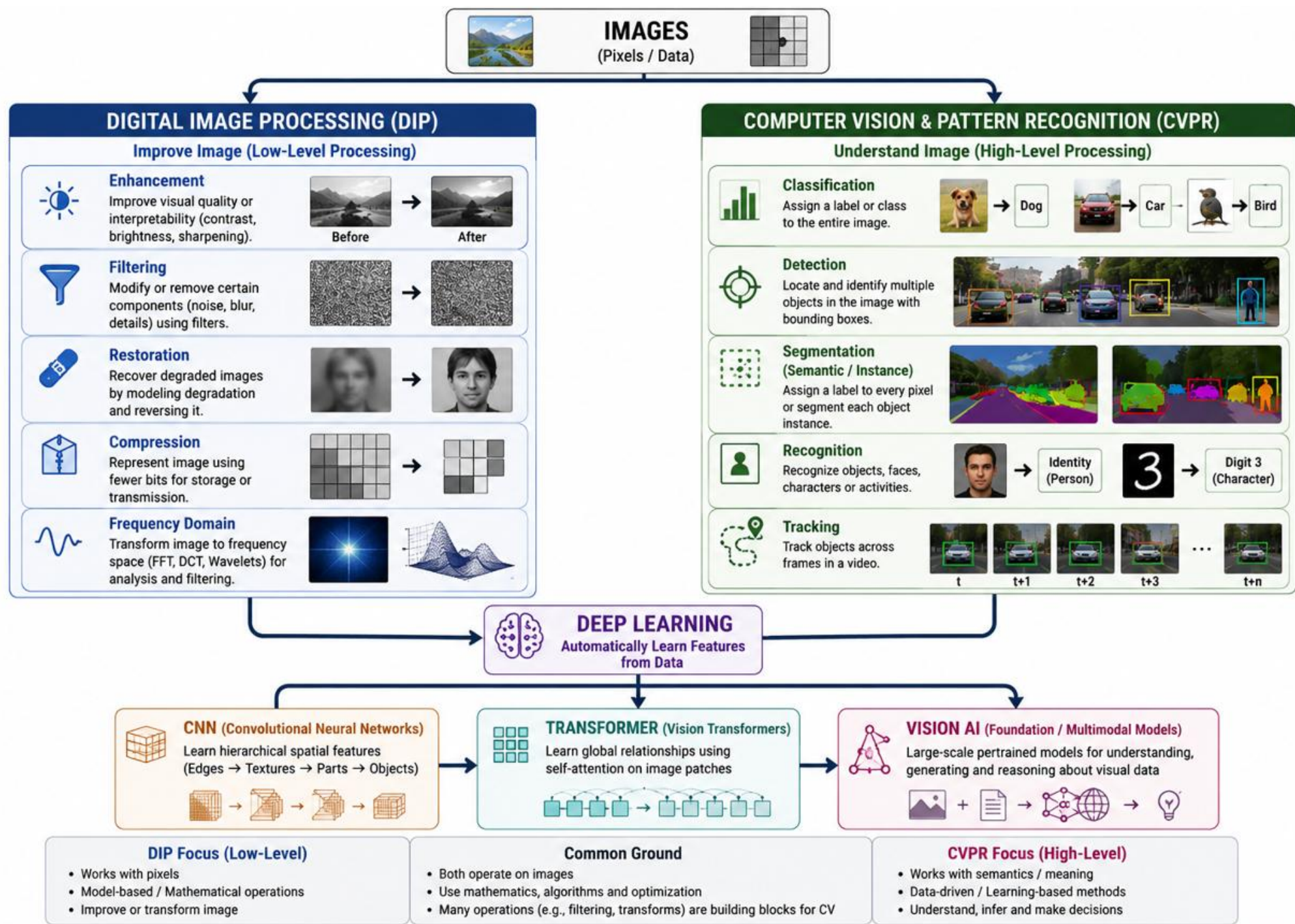
Application	Image Processing Task
WhatsApp photo	Compression
Photoshop	Enhancement
CCTV	Noise removal
MRI	Contrast enhancement
Google Maps	Image stitching
Drone mapping	Orthomosaic generation
Fingerprint scanner	Enhancement
Face filter	Image transformation

EVOLUTION OF IMAGE PROCESSING



CHRONOLOGY OF IMAGE FORMATION (SYMBOLIC PIPELINE)

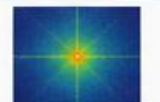
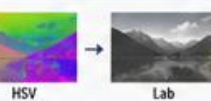




DIGITAL IMAGE PROCESSING (DIP) vs. COMPUTER VISION & PATTERN RECOGNITION (CVPR)

DIP improves or transforms images.

CVPR understands and extracts meaning from images.

TERM	DIGITAL IMAGE PROCESSING (DIP) – Low-Level	
	Goal: Improve image quality, extract useful information, or transform representation.	
IMAGE ACQUISITION 	Capturing image from a sensor and converting real world scene into digital form.	 →  →  Real World Sampling Digital Image
ENHANCEMENT 	Improve visual appearance or interpretability. (e.g., contrast, sharpness, brightness)	 →  Before After Enhancement
FILTERING 	Modify or remove certain components (noise, blur, details) using filters.	 →  Noisy Image Filtered Image
HISTOGRAM 	Shows distribution of pixel intensities. Used for analysis and enhancement.	 →  Frequency vs. Intensity (0 to 255)
TRANSFORMATIONS (FREQUENCY DOMAIN) 	Convert image to another domain (e.g., Fourier, DCT, Wavelet) for analysis or processing.	 → 
COLOR IMAGE PROCESSING 	Process and convert images in different color spaces (RGB, HSV, Lab, YCbCr).	 →  →  RGB HSV Lab
MORPHOLOGICAL OPERATIONS 	Process shapes using structuring elements. (e.g., erosion, dilation, opening, closing)	 →  →  →  Original Dilation Erosion Opening
SEGMENTATION (Traditional) 	Partition image into meaningful regions based on intensity, edges, or texture.	 →  Original Image Segmented Regions
COMPRESSION 	Represent image using fewer bits for storage or transmission. (e.g., JPEG, JPEG2000)	 →  Original (High Bits) Compressed (Low Bits)
RESTORATION 	Recover image degraded by noise or blur using mathematical models.	 →  Blurred/Noisy Image Restored Image

Typical Tools / Techniques (DIP)

Matlab, OpenCV, scikit-image, ImageJ, FFT, DCT, Wavelet, Morphology, Filters, Histograms

Prepares Images
Understands Images

COMMON FOUNDATION

-  Image Representation (Pixels, Resolution)
-  Sampling & Quantization
-  Histogram
-  Color Models
-  Transformations

INPUT FOR CVPR

Enhanced, Filtered, Transformed Images

DIP + CVPR = COMPLETE VISION SYSTEM
DIP enhances the signal.
CVPR extracts the meaning.

COMPUTER VISION & PATTERN RECOGNITION (CVPR) – High-Level		TERM
Goal: Extract semantic information, recognize objects and scenes, and make decisions.		
 Cat Dog Bird Car	CLASSIFICATION  Assign a label or class to the entire image. (e.g., cat, dog, car)	SYMBOLIC / MEANING
	DETECTION  Locate and identify multiple objects in the image with bounding boxes.	Locate and identify the lineage with m bounding boxes.
	SEGMENTATION (Semantic)  Assign a class label to every pixel. (e.g., road, building, tree)	Assign a class label to every pixel. (e.g., road, building, tree)
	SEGMENTATION (Instance)  Detect and segment each object instance separately.	Detect and segment each object instance separately.
	FEATURE / KEYPOINT DETECTION  Detect important points or features (corners, blobs) in the image.	Detect important points or features (corners, blobs) in the image.
 t t+1 t+2 t+n	TRACKING  Track object(s) across frames in video.	Track object(s) across frames in video.
 Road, Sidewalk, Building, Tree, Car, Person	SCENE UNDERSTANDING  Understand entire scene: objects, relationships, context.	Understand entire scene: objects, relationships, context.
 Walking Running Jumping Clapping	ACTION RECOGNITION  Recognize actions or activities in video.	Recognize actions or activities in video.
	POSE / 3D ESTIMATION  Estimate 2D/3D pose or structure of objects/people.	Estimate 2D/3D pose or structure of objects/people.
 A road near the lake with mountains and trees.	IMAGE CAPTIONING / RECOGNITION  Generate natural language description of the image.	Generate natural language description of the image.

Typical Tools / Techniques (CVPR)

OpenCV, PyTorch, TensorFlow, CNN, R-CNN, YOLO, U-Net, Transformers, Tracking, SLAM, etc.

Suggested Reference List for the Syllabus (Download the coloured references)

1. **R. C. Gonzalez and R. E. Woods**, *Digital Image Processing*, 4th Edition, Pearson. (Primary Text)
2. **R. C. Gonzalez, R. E. Woods, and S. L. Eddins**, *Digital Image Processing Using MATLAB*, 3rd Edition.
3. **Richard Szeliski**, *Computer Vision: Algorithms and Applications* (Free online).
4. **Wilhelm Burger and Mark Burge**, *Digital Image Processing: An Algorithmic Introduction using Java*.
5. **Anil K. Jain**, *Fundamentals of Digital Image Processing*.
6. **OpenCV Documentation**.
7. **MATLAB Image Processing Toolbox Documentation**.
8. **scikit-image Documentation**.
9. **PyTorch Tutorials**.
10. **Ultralytics YOLO Documentation**.

This combination gives students a complete learning path:

- **Gonzalez & Woods** for theory and mathematics,
- **Digital Image Processing Using MATLAB** for laboratory implementation,
- **Szeliski** for modern computer vision concepts,
- **PyTorch/YOLO/Hugging Face** for the final deep learning and transformer modules.